

The ANNI phase diagram under CNOT-target depolarizing noise

Q-SITE Hacks 2026 • Scientific Open Challenge

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Code, data and notebook: github.com/BenjaminKaganovitch/qsites-2026-open-challenge. Every number below is taken from the saved analysis outputs in that repository. AI coding assistants were used, as permitted by the organisers.

1 Problem and conventions

1.1 Model and diagnostics. We study $H = -\sum_i Z_i Z_{i+1} + \kappa \sum_i Z_i Z_{i+2} - h \sum_i X_i$ with Pauli operators, $J_1 = 1$ and **periodic rings**, for $\kappa \in [0, 1]$ and $h \in [0, 2]$. The phases are identified from longitudinal correlations, not from one-point magnetisations, because the latter vanish in finite symmetric ground states [S07]. We use $S(q) = \frac{1}{N} \sum_{ij} e^{iq(i-j)} \langle Z_i Z_j \rangle$ with the self-terms kept, and the floor-subtracted order parameters $O_F = (S(0) - 1)/(N - 1)$ and $O_A = (S(\pi/2) - 1)/(N/2 - 1)$, which equal 1 for perfect order and 0 for an uncorrelated state. We also use the transverse magnetisation m_x . Validated: $S(0) = N$ (ferro), $S(\pi/2) = N/2$ (antiphase), and $S(q) = 1$ for both $|+\rangle^N$ and $I/2^N$, which m_x separates.

1.2 Reference curves. The overlays are approximate thermodynamic references, not finite-ring labels [S10, S13]: the second-order Ising estimate (rationalised so that $h_I(0) = 1$), the lower antiphase-floating fit $1.05(\kappa - \frac{1}{2})$, which is a **Pokrovsky–Talapov** transition [S12], and the upper **KT** fit $1.05\sqrt{(\kappa - \frac{1}{2})(\kappa - 0.1)}$.

2 Methods

2.1 Exact reference. SciPy’s sparse Lanczos solver returns the lowest six eigenpairs on 31×31 grids for $N = 8, 10, 12$, and on fine cuts (21 values of $\kappa \times 401$ values of h) for all three sizes. Warm-started h sweeps agree with independent cold starts to 10^{-13} ; degenerate ground spaces are treated as equal mixtures, and fidelity is measured between ground subspaces. The Hamiltonian matches the starter exact diagonalisation and PennyLane’s matrix representation to 7×10^{-15} .

2.2 Classifier. The thresholds come from anchor regions fixed by limiting-case physics, not from the overlay curves: ferro at $\kappa \leq 0.2$ and $h \leq 0.2$, antiphase at $\kappa \geq 0.8$ and $h \leq 0.1$, paramagnet at $h \geq 1.8$. This gives $T = 0.541$ for the order parameters and $T_x = 0.490$ for m_x . A point is ferro or antiphase if $\max(O_F, O_A) \geq T$, paramagnet if not but $m_x \geq T_x$, and otherwise *unpolarised*. Labels that flip for thresholds ± 0.1 are marked ambiguous. We also report a threshold-free *relative-dominance* label, $\arg \max(O_F, O_A, m_x^2)$. Under uniform terminal depolarisation all three quantities contract by the same factor a^{2m} with $a = 1 - 4p/3$ [S28], so this label is noise-invariant in that benchmark; any change it shows therefore reveals circuit-specific noise action.

2.3 Circuit. We use a Hamiltonian-variational ansatz [S30]: $|+\rangle^N$ followed by $L = N/2 = 4$ layers of $e^{-ib} \sum X e^{-ic} \sum ZZ_{nnn} e^{-ia} \sum ZZ_{nn}$, which gives 12 parameters and preserves translation and parity symmetry. Each ZZ bond compiles to CNOT–RZ–CNOT, for **128 CNOTs** in total. (A hardware-efficient RY/CNOT ring stalled at 1–3% energy error near the boundaries even at depth 6.) Parameters are optimised at $p = 0$ by L-BFGS with exact adjoint gradients, forward and reverse warm-started sweeps and random restarts. Noise is PennyLane’s DepolarizingChannel(p) on the target after every CNOT on the default.mixed simulator, with the 128 channels verified on the circuit tape; a NumPy density-matrix engine matching default.mixed to 3×10^{-14} runs the zero-noise-extrapolation and response sweeps.

2.4 Protocol. Fixed parameters: the same parameters, circuit, grid, thresholds and colour scales at $p = 0, 0.01, 0.05$. The shift $\delta h(\kappa, p) = h_b(p) - h_b(0)$ is taken for the *same circuit*; the preparation bias $h_b(0) - h_b^{\text{ED}}$ is reported separately.

3 Clean results and finite size

3.1 Circuit accuracy. At $p = 0$ the circuit is an accurate stand-in for the exact ground states. The median relative energy error is 4×10^{-5} and the median $1 - F$ is 5×10^{-5} (with F the ground-subspace fidelity). Only 1.5% of points, all around the multicritical point $\kappa \approx 0.5$, have $F < 0.9$, and the circuit labels agree with exact diagonalisation (ED) at 98.9% of points (Figure 1). The preparation bias on the threshold boundary is at most 0.034 outside $0.42 < \kappa < 0.6$, and 0.066 at $\kappa = 0.57$ next to the multicritical point. Two independent optimiser seeds agree on it to a median of 0.007 (maximum 0.05).

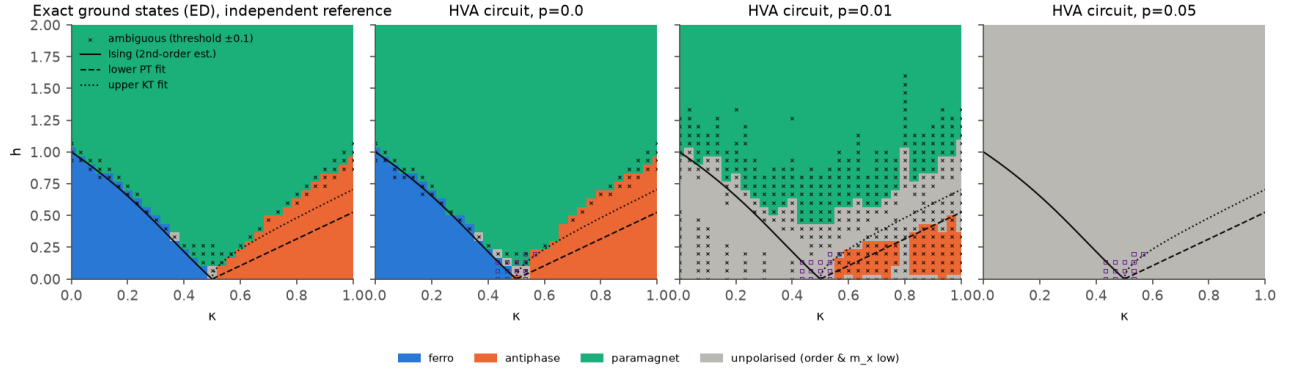


Figure 1. Phase diagrams on the 31×31 grid ($N = 8$, periodic). Left to right: exact ground states (ED), then the HVA circuit at $p = 0, 0.01$ and 0.05 with fixed parameters and fixed thresholds. Purple squares mark clean preparation fidelity below 0.9; $\times$ marks threshold-ambiguous points. Curves: Ising estimate (solid), lower PT fit (dashed), upper KT fit (dotted).

Table 1. Boundary positions and noise-induced shifts for the HVA circuit. h_T is the fixed-threshold crossing and h_D the maximum of $-dO/dh$. The grid step is 0.067; the seed spread in δh_D at $p = 0.01$ has median 0.011 and maximum 0.038. The full table for all 31 values of κ is in the repository.

κ	Order	h_T^{ED}	Prep. bias	$h_D(0)$	$\delta h_D(p=0.01)$	$\delta h_D(p=0.05)$	h_T at $p=0.01$
0.2	ferro	0.685	-0.000	0.653	+0.027	+0.64 [†]	absent
0.3	ferro	0.487	-0.016	0.469	+0.015	+0.19	absent
0.4	ferro	0.246	+0.000	0.248	+0.003	+0.006	absent
0.7	anti	0.467	+0.018	0.477	-0.001	-0.004	absent
0.8	anti	0.650	+0.024	0.672	+0.003	+0.045	absent
0.9	anti	0.809	+0.004	0.723	+0.000	+0.098	absent

[†] Unreliable: the order signal at $p = 0.05$ is too small to locate a derivative peak.

3.2 Finite-size behaviour. The **ferro-para** boundary is almost independent of size and tracks the Ising estimate. At $\kappa = 0.2$, the fidelity-susceptibility peak moves from 0.616 ($N = 8$) to 0.629 ($N = 12$), towards the DMRG value 0.639 reported in [S10]; the second-order estimate is 0.653. The **antiphase** boundary drifts strongly with size. At $\kappa = 0.75$ the order crossing falls from 0.562 ($N = 8$) to 0.432 ($N = 12$), approaching the KT fit at 0.423 from above. $N = 10$ cannot hold a period-4 pattern and is visibly anomalous: at $\kappa = 0.6$ its crossing is 0.397, against 0.244 ($N = 8$) and 0.164 ($N = 12$). That is a commensurability effect, not physics of the thermodynamic limit.

3.3 Floating phase: unresolved. At $N = 12$ the half-chain entropy of the pure ground state peaks close to the KT fit ($\kappa = 0.75$ and $\kappa = 0.9$), and the boundary drifts towards the KT line as N grows. Both are *compatible* with a critical region. However, the dominant q^* stays locked at $\pi/2$ between the PT and KT fits at every size. The only jumps in q^* occur above the KT line, inside the modulated paramagnet [S10], and the momentum resolution ($\Delta q = \pi/6$ at $N = 12$) cannot resolve incommensurability. We therefore claim no floating phase [S12, S23]. Label-free k-means on $(S(q)/N, m_x)$ matches the clean rule labels at 99.1% (consistency, not independent validation).

4 Noise: effects, magnitude and mechanism

4.1 Noise level $p = 0.01$. The order parameters contract to a median of 0.60 of their clean values, against 0.65 if every target hit acted as terminal noise. The contraction depends on the phase. At $h \rightarrow 0$, O_F at $\kappa = 0.2$ drops from 1.00 to 0.37, while O_A at $\kappa = 0.9$ drops only to 0.59. Under the fixed threshold, **the ferromagnet disappears entirely** (0% of ED-ferro points keep their label), 46% of antiphase points and 89% of paramagnet points survive. Where a threshold crossing survives, it moves **to lower h** by 0.04–0.48, the same sign as the exact first-order prediction $-p \partial_p O / \partial_h O$ but 1.5–4 $\times$ larger, because the response is already nonlinear. The **derivative-peak boundaries barely move** (Table 1): the median $|\delta h|$ is 0.005 and 28 of 31 rows stay within 0.03 (half a grid step). The exceptions are $\kappa = 0.13$ and 0.27 (+0.04) and $\kappa = 0.93$ (-0.12). That is what [S28] predicts for a near-uniform rescaling, and the relative-dominance map agrees with the clean-ED relative-dominance map at 97% of points (99% at $p = 0$).

4.2 Noise level $p = 0.05$. The state is essentially maximally mixed: the median purity is 0.0047, against 0.0039 for $I/2^8$. All points are unpolarised under fixed thresholds; in the relative-dominance map the *paramagnet* shrinks: m_x keeps 16% of its clean value, so m_x^2 keeps about 2.6%, while the order parameters keep a median of 8.5%.

4.3 Mechanism from exact channel-insertion sums. We evaluated the first-order response $\partial_p \langle O \rangle = \sum_g \text{Tr}\{O_g \mathcal{L}_t(\rho_g)\}$ location by location [S28]. For the ferro state $d \ln O_F / dp = -99$, for the antiphase $d \ln O_A / dp = -69$, and for m_x in the paramagnet -32 to -48 . The terminal-equivalent value is -43 for pair correlators. Mid-circuit errors, carried through later CNOT-RZ-CNOT gadgets, damage ZZ order more than end-of-circuit errors. Early layers contribute most: for the ferro point, dO_F / dp is -27 from layer 1 and -17 from layer 4. One consistent reading of why the ferromagnet is most fragile: its GHZ-like finite-ring ground state loses all $N(N-1)$ positive correlators in O_F to a single propagated X/Y error, whereas O_A mixes signs and the damage partly cancels. This ranking is specific to this circuit and diagnostic, and reverses the organiser’s hypothesis [S01]. The noise does not change the Hamiltonian’s spectrum, so these are maps of prepared-state diagnostics.

4.4 Depth trade-off. We repeated the fixed-parameter experiment for $L = 1-4$ on four κ cuts. The clean error $\langle |O - O^{\text{ED}}| \rangle_h$ falls with depth, while the noise error grows with it, so the best depth shrinks as p grows. The antiphase is already accurate at $L = 2$ (clean error 0.024 at $\kappa = 0.8$), and $L = 2$ is the best depth there at both $p = 0.01$ and $p = 0.05$. The ferromagnet needs $L = 4$ to be accurate (error at $L = 2$: 0.17), and at $p = 0.01$ its best depth is $L = 3$. So the ferro state needs the full depth and carries the most channels.

5 Error mitigation

5.1 Zero-noise extrapolation. We ran Richardson extrapolation on every C_{ij} and $\langle X_i \rangle$ at λp for $\lambda = 1, 2, 3$ (linear scaling), with the $p = 0$ truth held out, and rebuilt $S(q)$ before classifying. At $p = 0.01$ this is effective: the median $|\Delta C_{ij}|$ falls from 0.063 to 0.011, $|\Delta m_x|$ from 0.26 to 0.023, and label agreement with the clean circuit rises from 70% to 98%. Threshold boundaries come back to within $|\delta h| \leq 0.09$, with a median of 0.03 (for example, $\kappa = 0.93$: $-0.48 \rightarrow -0.04$). The cost is a variance-amplification factor $\sum \gamma^2 = 19$. At $p = 0.05$ zero-noise extrapolation fails for both linear and composed-channel scaling: the signal is about 1% and the extrapolation reaches $p = 0.15$.

5.2 Parity verification. Verification with $P = \prod X_i$ accepts 71% of the weight at $p = 0.01$ and 51% at $p = 0.05$. It lowers the C_{ij} error only from 0.063 to 0.053, because most errors stay inside the even-parity sector.

5.3 Noise-aware re-optimisation. As a separate protocol ($\kappa = 0.2, 0.3, 0.7, 0.8$; 8 values of h each), minimising $\text{Tr}(H\rho_p)$ with exact gradients through the channels lowers the noisy energy by 0.2–1.0 per cut, but the order parameters move by only -0.016 to $+0.028$, and their error against ED is unchanged (for example, $\kappa = 0.8$ at $p = 0.01$: $0.225 \rightarrow 0.198$). The optimiser gains energy mainly through m_x ($+0.03$ to $+0.09$) by leaving the ground state: at $\kappa = 0.2, h = 0$ the clean energy of the re-optimised parameters is -5.83 , against -6.40 for the true ground state. Energy-optimal noisy states are therefore not diagnostic-optimal.

6 Conclusions and limitations

The clean diagram is reproduced accurately on 8–12-site rings. For a symmetric circuit accurate enough to represent these ground states (128 CNOTs), $p = 0.01$ **already removes the ferromagnet under fixed thresholds while boundary positions defined by derivatives stay put**; $p = 0.05$ **leaves no usable signal**. Zero-noise extrapolation recovers the $p = 0.01$ map almost fully. Limitations: small N ; overlays are approximate; the $\kappa \approx 0.5$ region is masked for preparation quality; the maps use the fixed-parameter protocol (re-optimisation only on cuts); simulations are exact, with no shot noise.

References. Full entries are in the repository’s literature notes. [S01] Organiser handout; [S07] starter-kit audit; [S10] Beccaria, Campostrini, Feo, PRB 73, 052402 (2006); [S12] Cea et al., arXiv:2402.11022 (2024); [S13] Beccaria, Campostrini, Feo, PRB 76, 094410 (2007); [S19] Temme, Bravyi, Gambetta, PRL 119, 180509 (2017); [S21] Bonet-Monroig et al., PRA 98, 062339 (2018); [S23] Sandvik, AIP Conf. Proc. 1297, 135 (2010); [S28] our channel-response derivation; [S30] Wecker, Hastings, Troyer, arXiv:1507.08969 (2015). Tools: PennyLane 0.44.1, NumPy, SciPy, Matplotlib.